

# Brandon Nguyen

Springfield, MA | brandon@umass.edu | 413-309-1636 | linkedin.com/in/brndngyn | U.S. Citizen

## Education

**University of Massachusetts, Amherst**

Expected May 2028

BS in Mechanical Engineering

GPA: 3.79

Courses: Machine Design, Thermodynamics, Fluid Mechanics, Dynamics, Manufacturing Processes, Materials Science

## Projects

**Physics-Based Airfoil Parameterization Validation**

June 2026 – Present

*UMass Amherst Wind Energy Center*

- Assembled and standardized a 42-airfoil coordinate database spanning multiple wind turbine airfoil families, enforcing uniform formatting and cosine point spacing and verifying thickness-to-chord ratios against published references
- Implemented a physics-based airfoil parameterization (PBAP) and benchmarked its shape-fitting accuracy against the industry-standard CST method across multiple parameter counts and basis functions, validating solution uniqueness through multi-start optimization
- Built an XFOIL simulation pipeline and ran 500+ tests across varying Reynolds numbers and surface conditions to evaluate whether shape-fitting accuracy preserves lift and drag performance

## Experience

**UMass Amherst** | Undergraduate Teaching Assistant

September 2026 – Present

*Amherst, MA*

- Design weekly homework problems for MIE 230: Thermodynamics I, targeting concepts students struggle with on exams and in office hours
- Lead weekly help sessions and exam review workshops for a 200+ student course, breaking down entropy, power and refrigeration cycles, and heat transfer into step-by-step problem-solving approaches
- Grade homework and exams for 200+ students, giving written feedback that pinpoints errors in setup, assumptions, and unit handling

**B&E** | HVAC Installation Technician

June 2025 – August 2026

*Springfield, MA*

- Completed ductless mini-split HVAC installations on 70+ residential jobs, installing 2–4-zone systems and cutting install time by up to 50% as lead installer
- Installed and routed condensate drain lines, incorporating condensate pumps where gravity drainage was not feasible, to prevent leaks and water backup
- Mounted and leveled 100+ condenser brackets and pads to exact measurements, ensuring structural stability and proper unit clearance
- Routed, cut, and connected refrigerant line sets; wired condensers from the electrical disconnect and ran communication wiring to indoor units per manufacturer specifications

**City Cleaners** | Cashier

May 2023 – August 2024

*Springfield, MA*

- Managed front-desk operations for a high-volume dry-cleaning business, processing customer orders and cash and card transactions
- Resolved customer service issues and coordinated special requests with production staff to meet customer deadlines

## Skills

**Software:** SolidWorks | ANSYS | MATLAB | Python | Autodesk Inventor | Fusion 360 | MS Excel

**Engineering:** CAD / 3D Modeling | CNC Machining | Manual Machining | GD&T | Prototyping